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AMZ 5450

HIGH PRECISION SMART PRESSURE TRANSMITTER WITH HART® COMMUNICATION

DESCRIPTION

The AMZ 5450 series pressure transmitters are designed for various industrial applications, offering an accuracy of up to ±0.075% of span and active temperature compensation that eliminates additional temperature-induced errors. They output a 4–20 mA two-wire analog signal with superimposed HART® digital communication. The local LCD display features a backlight and can be mounted at 0°, 90°, 180°, or 270° for flexible installation.

The transmitters feature high overload resistance, excellent long-term stability, and a wide range of process connections and options.

SPECIFICATIONS

Pressure ranges: 0...15 mbar to 0...600 bar

Overpressure: up to 1050 bar

Accuracy: up to ±0.075% of span

Output / Communication: 4...20 mA / HART®

Explosion protection: 0Ex ia IIC T6...T4 Ga X, 1Ex d IIC T6...T4 Gb X, 1Ex d ia IIC T6...T4 Gb X

Diaphragm: stainless steel

Process connections:
- Version 00 - 1/2" NPT internal - standard, other configurations provided by adapter;
- Version 0T - connection configuration to be agreed upon order

Turndown: up to 100:1

LCD display with backlight

Hydrostatic version

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APPLICATIONS

Liquid, steam and gas pressure measurement

Level monitoring in open tanks

TECHNICAL SPECIFICATIONS

MEASURING RANGES

Version 00			Version 0T		
Gauge pressure range, P_N*	**Turndown ratio, P_N/P_{set}***	**Overpressure, MPa**	**Gauge/Absolute pressure range, P_N***	**Turndown ratio P_N/P_{set}***	**Overpressure, MPa**
0...1.5 kPa	10:1	1	–	–	–
0...7.5 kPa	30:1	4	–	–	–
0...37 kPa	100:1	13.8	0...37 kPa	10:1	0.1
0...187 kPa	100:1	13.8	0...187 kPa	10:1	0.6
0...690 kPa	100:1	13.8	0...690 kPa	10:1	1.5
0...2 MPa	100:1	13.8	0...2 MPa	10:1	6
0...7 MPa	100:1	13.8	0...7 MPa	10:1	10
–	–	–	0...10 MPa	10:1	15
–	–	–	0...20 MPa	10:1	30
–	–	–	0...40 MPa	10:1	105
–	–	–	0...60 MPa	10:1	105

* – By default, the nominal measurement range P_N equals the upper range limit (URL), with the lower range limit (LRL) set to 0. For gauge pressure transmitters, the LRL can be set to a value equal to or less than the URL in magnitude (but negative), or to -100 kPa if the URL is ≥ 187 kPa.
The calibrated span (P_{set}) is defined as the difference between URL and LRL.
The pressure transmitter supports the following measurement units: inH_2O, inHg, ftH_2O, mmH_2O, mmHg, psi, bar, mbar, g/cm^2, kg/cm^2, Pa, kPa, MPa, atm, Torr. Switching between measurement units can be performed remotely using a HART® modem/communicator or locally.
When switching units, the range of the digital display range should be taken into account.

PERFORMANCE

Pressure ranges	**Turndown**	**Accuracy, % of span***	
		Version 00	**Version 0T**
P_N = 1.5 kPa	$P_N/P_{set} \le 2$	±0.1	–
	$2 < P_N/P_{set} \le 10$	$\pm[0.02 \cdot (P_N/P_{set}) + 0.06]$	–
7.5 kPa ≤ P_N ≤ 60 MPa	$P_N/P_{set} \le 10$	±0,075	$\pm[0.02 \cdot (P_N/P_{set}) + 0.08]$
	$10 < P_N/P_{set} \le 40$	$\pm[0.00375 \cdot (P_N/P_{set}) + 0.0375]$	–
	$40 < P_N/P_{set} \le 100$	$\pm[0.00465 \cdot (P_N/P_{set}) + 0.0015]$	–

Pressure ranges	**Turndown**	**Temperature effect, % of span / 10 °C**		**Long Term Stability**	
		Version 00	**Version 0T**	**Version 00**	**Version 0T**
P_N = 1.5 kPa	$P_N/P_{set} \le 2$	$\pm[0.075 \cdot (P_N/P_{set}) + 0.025]$	–	±0.2% of span / year	
	$2 < P_N/P_{set} \le 10$	$\pm[0.050 \cdot (P_N/P_{set}) + 0.075]$	–		
P_N = 7.5 kPa	$P_N/P_{set} \le 5$	$\pm[0.040 \cdot (P_N/P_{set}) + 0.025]$	–		
	$5 < P_N/P_{set} \le 40$	$\pm[0.030 \cdot (P_N/P_{set}) + 0.075]$	–		
37 kPa ≤ P_N ≤ 60 MPa	$P_N/P_{set} \le 5$	$\pm[0.010 \cdot (P_N/P_{set}) + 0.030]$	$[0.02 \cdot (P_N/P_{set})]$	±0.15% of span / 5 year	±0.1 % of span / year
	$5 < P_N/P_{set} \le 100$	$\pm[0.012 \cdot (P_N/P_{set}) + 0.023]$	$[0.02 \cdot (P_N/P_{set})]$		

** – Accuracy includes non-linearity, hysteresis and non-repeatability.

Compensated temperature range	-20...+80 °C; -40...+60 °C (optional)
Power supply effect (Nominal power supply: 24 V ±10%)	≤ ±0.05% of span / 10 V
Load resistance effect	≤ ±0.05% of span / kΩ
Response time (10...90%)	< 200 ms

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OPERATING CONDITIONS

Medium temperature*	-50...+105 °C (Version 00) or -50...+125 °C (Version 0T)					
Ambient temperature	-50...+85 °C (refer to the temperature class limits for Ex versions)					
Storage temperature	-40...+85 °C					
Approvals	1Ex d IIC T6…T4 Gb X, 1Ex d ia IIC T6...T4 Gb X			0Ex ia IIC T6…T4 Ga X		
Temperature class	T4	T5	T6	T4	T5	T6
Ambient temperature	-50…+85 °C	-50…+70 °C	-50…+60 °C	-50…+80 °C	-50…+60 °C	-50…+50 °C
Vibration resistance	10 - 60 Hz, 0.21 mm peak to peak displacement / 60 - 2000 Hz, 3g					
Shock resistance	100 g / 11 ms					
Sensor service life	> 100 × 10^6 cycles					

* – Temperature limitations determined by the sensor design and materials, including the sensing element, are considered the ultimate limits for the entire device, encompassing seals, parts, and components in contact with the measured medium

MECHANICAL SPECIFICATIONS

Housing material	standard - aluminum alloy, optional - stainless steel 316L (1.4404)
Pressure port (fitting) material	stainless steel 316L (1.4404)
Seal (for adapters with DIN 3852)	EPDM (-50…+125 °C); FKM (-25…+125 °C); NBR (-25…+100 °C)
Diaphragm	stainless steel 316L (1.4435)
Mounting kit, Mounting bracket	carbon steel, stainless steel
Display protective cover	polycarbonate
Wetted parts	diaphragm, pressure port, seal
Process connections	Version 00 - 1/2" NPT internal. - Standard, other configurations are provided by an adapter, Version 0T - Can be agreed upon when ordering
Electrical connections	Cable entry configuration (housing opening/thread): 1/2" NPT or M20×1.5. Electrical connection configurations are presented in Appendix A**
Ingress protection	IP68
Dimensions, mm, max	Depends on the current configuration
Weight, kg, max	3,5
Design	General industry; Intrinsically safe 0Ex ia IIC T6...T4 Ga X; Flameproof enclosure 1Ex d IIC T6…T4 Gb X, 1Ex d ia IIC T6...T4 Gb X. The design allows for local configuration using a magnetic tool or configuration buttons in a hazardous area.

** – The number of possible electrical connection configurations is not limited to the list presented in Appendix A; the required configuration can be agreed upon at the time of ordering.

DIGITAL DISPLAY (optional)

Display	Value
Display digits	-1999…+9999
Display accuracy	± 0.1 % of span ± 1 digit

ELECTRICAL SPECIFICATIONS

Output signal	Power supply	Load resistance	Power consumption
4...20 mA / HART®	9…44 V (DC)	$\le [(U_S - U_{S\_Min}) / 0.02\ A]\ \Omega$***	≤ 21 mA

Minimum value of the supply voltage	Without HART®, U_{S_Min}	With HART®, $U_{S\_Min\_HART}$
With backlight off	9 V	14 V
With backlight on	12 V	17 V

*** – Maximum load resistance value depends on the supply voltage and the minimum supply voltage.
For the sensor to operate using the HART® protocol, the load resistance must be at least 250 ohms.
HART® is a registered trademark of the HART Communication Foundation.

Safe values for intrinsically safe design 0Ex ia IIC T6…T4 Ga X:

Parameter	2-wire
Maximum input voltage, Ui	28 V
Maximum input current, Ii	93 mA
Maximum input power, Pi	660 mW
Maximum internal inductance, Li	5 µH
Maximum internal capacitance, Ci	10 nF

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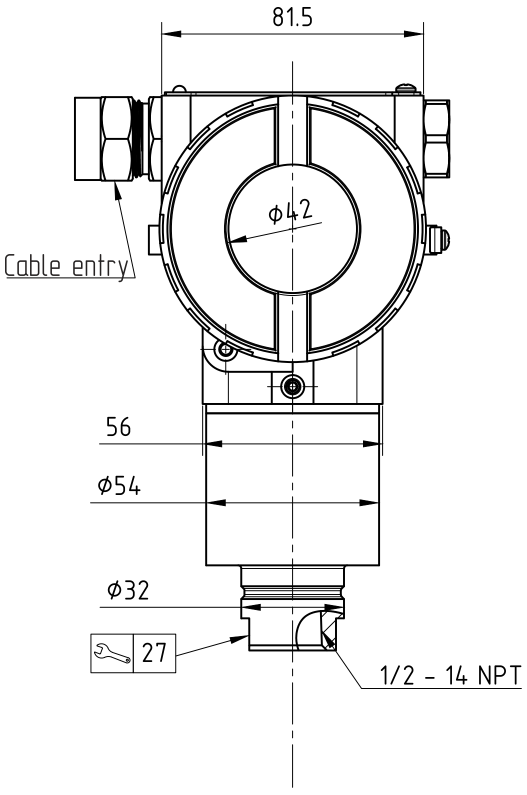
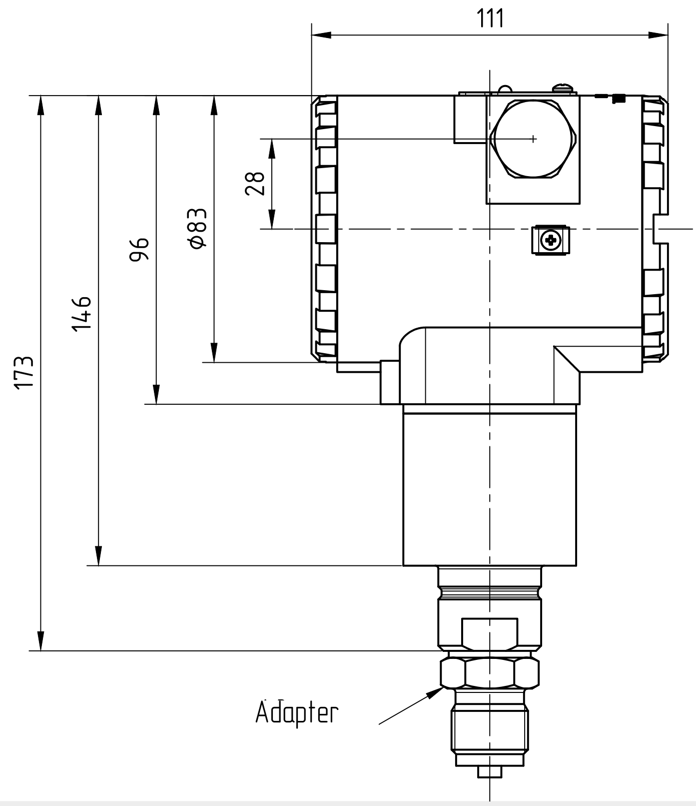
DIMENSIONS (mm) ALUMINUM ALLOY HOUSING

AMZ 5450 Version 00

Standard -1/2" NPT internal	Adapter M20x1.5 / G1/2" EN 837

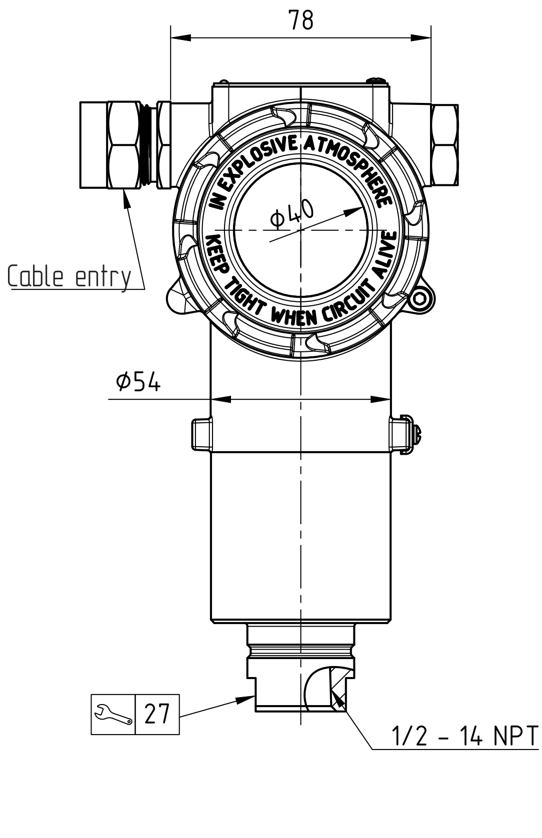


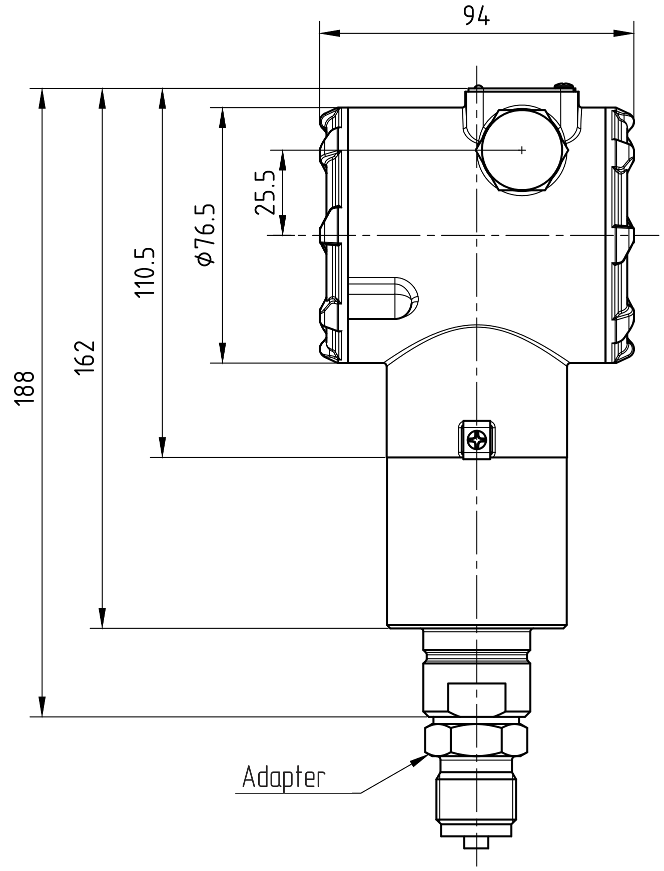


DIMENSIONS (mm) STAINLESS STEEL HOUSING

AMZ 5450 version 00-SS

Standard - 1/2" NPT internal	Adapter M20x1.5 / G1/2" EN 837





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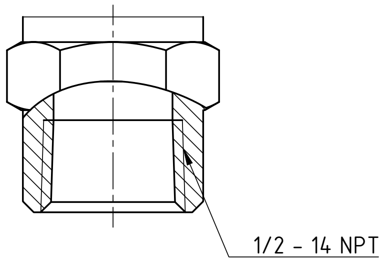
MECHANICAL CONNECTIONS FOR AMZ 5450, VERSION 00, DIMENSIONS (mm)

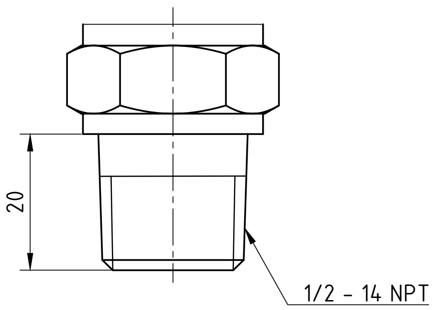
Standard mechanical connection

1/2" NPT; code 2

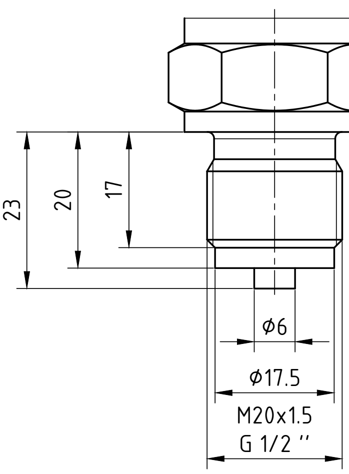


Adapter Configurations

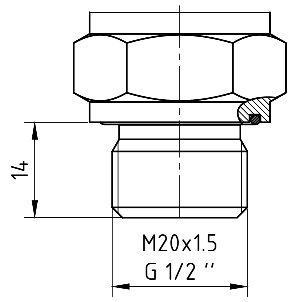
\1/2" NPT; code1



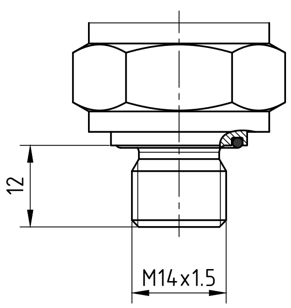
M20x1.5 EN 837; code 5
G1/2" EN 837; code 7



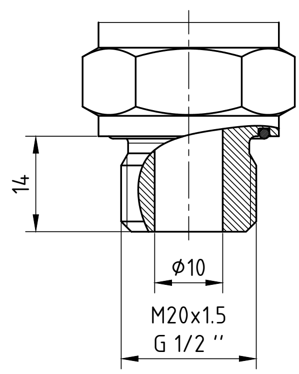
M20x1.5 DIN 3852; code 6
G1/2" DIN 3852; code 8



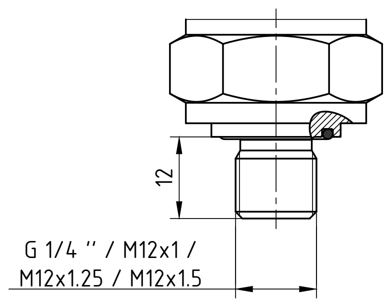
M14x1.5 DIN 3852; code 140



M20x1.5 DIN 3852 open port; code 206
G1/2" DIN 3852 open port; code 726



G1/4" DIN 3852; code 740
M12x1 DIN 3852; code 120
M12x1.25 DIN 3852; code 127
M12x1.5 DIN 3852; code 122



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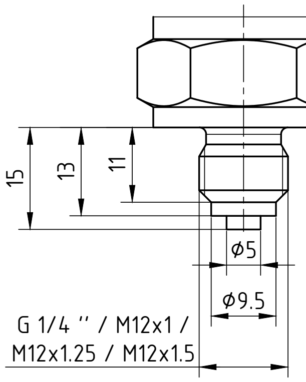
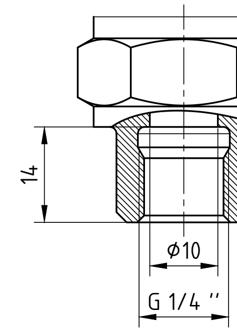
MECHANICAL CONNECTIONS FOR AMZ 5450, VERSION 00, DIMENSIONS (mm)

Adapter Configurations

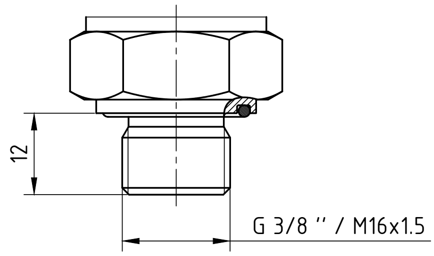
G1/4" EN 837; code 741
M12x1 EN 837; code 121
M12x1.25 EN 837; code 128
M12x1.5 EN 837; code 123



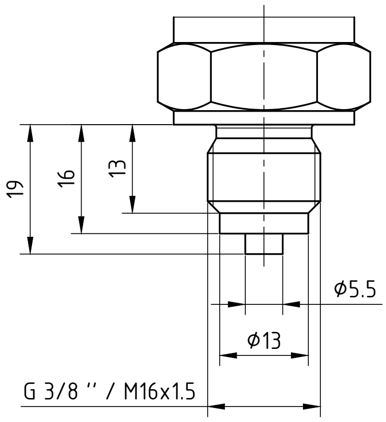
G1/4" internal thread,
through bore 10 mm; code 742



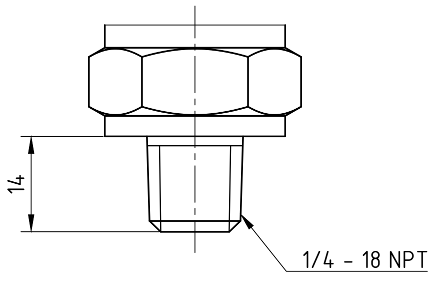
G3/8" DIN 3852; code 790
M16x1.5 DIN 3852; code 160



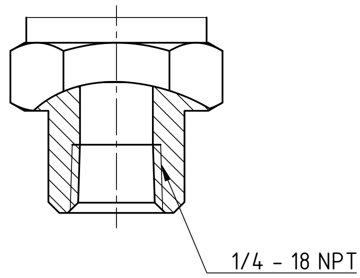
G3/8" EN 837; code 791
M16x1.5 EN 837; code 61



1/4" NPT; code 840



1/4" NPT internal; code 841



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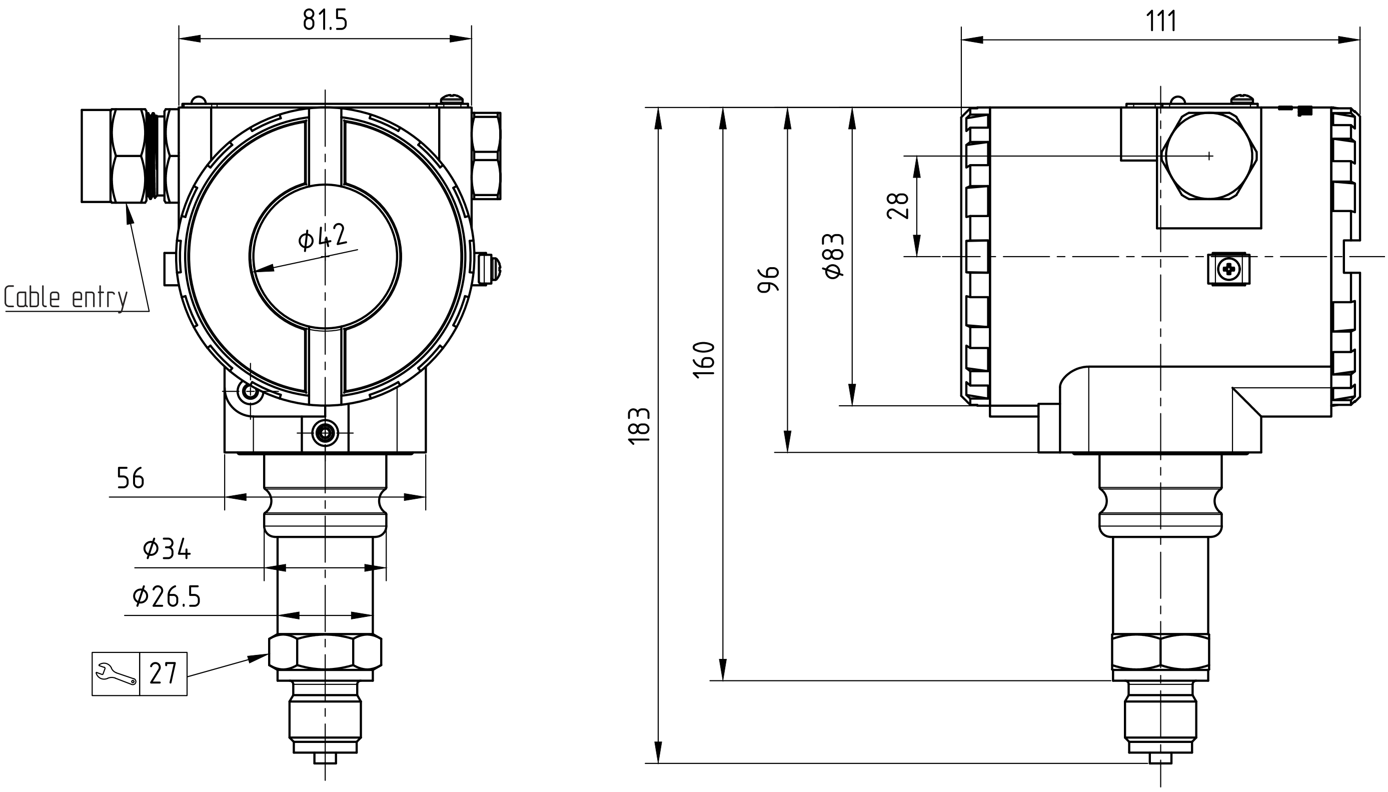
DIMENSIONS (mm) ALUMINUM ALLOY HOUSING

AMZ 5450 Version 0T

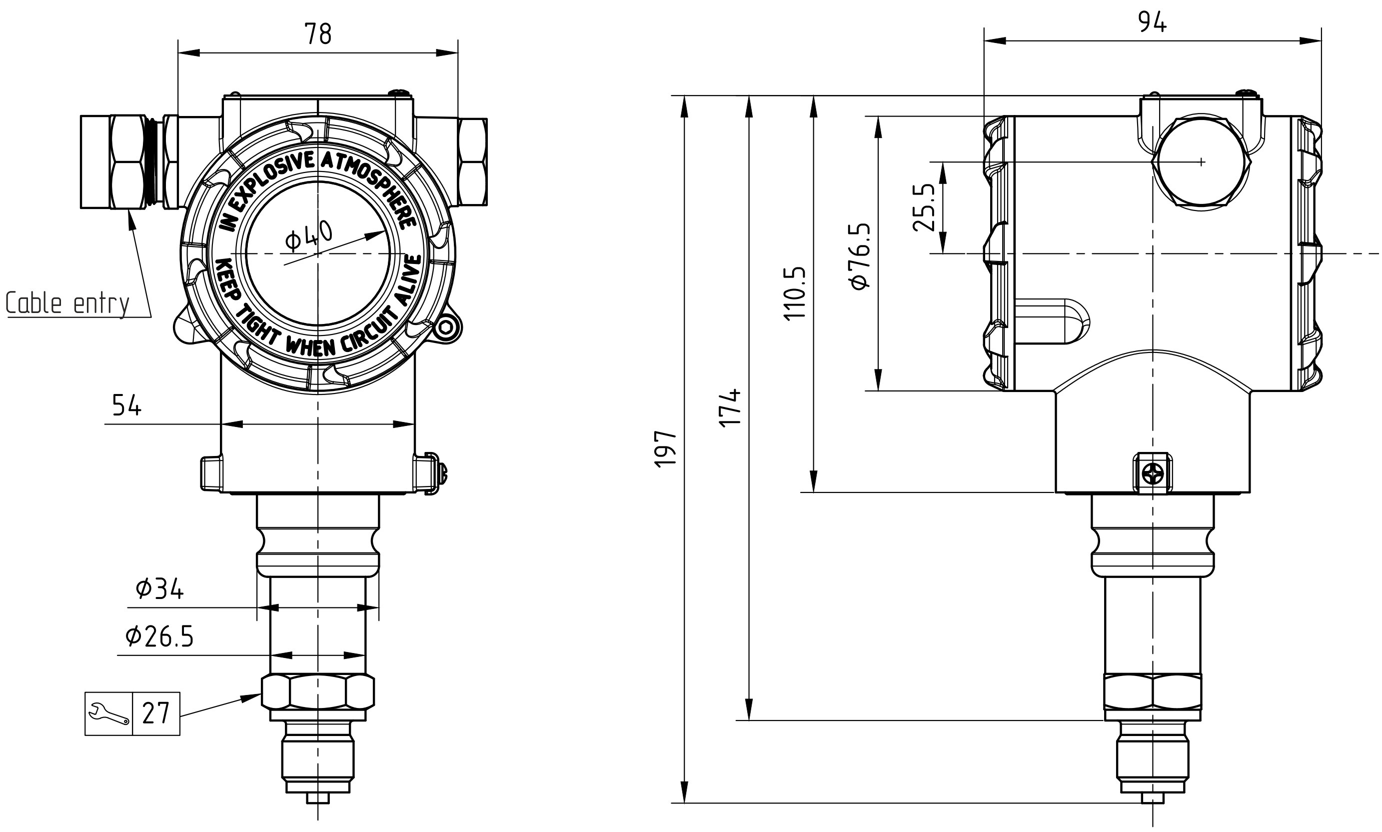
With pressure port G1/2", M20x1.5 EN 837



DIMENSIONS (mm) STAINLESS STEEL HOUSING

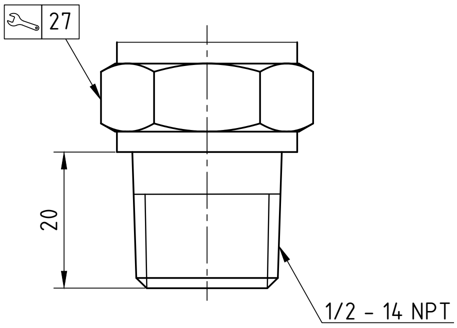
AMZ 5450 Version 0T-SS

With pressure port G1/2», M20x1.5 EN 837

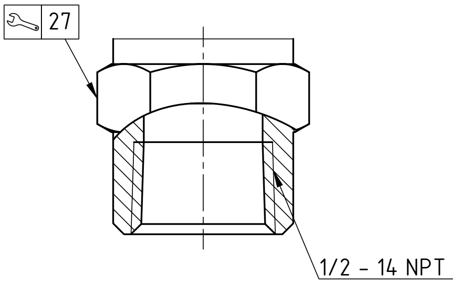


MECHANICAL CONNECTIONS FOR AMZ 5450 VERSION 0T, DIMENSIONS (mm)

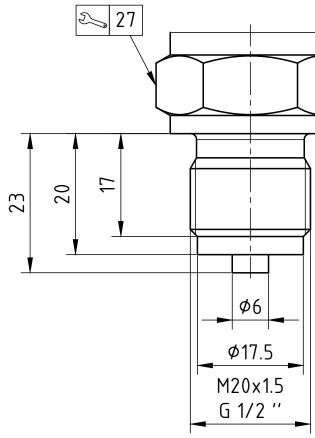
1/2" NPT; code 1



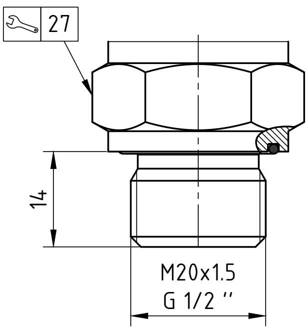
1/2" – 14 NPT internal; code 2



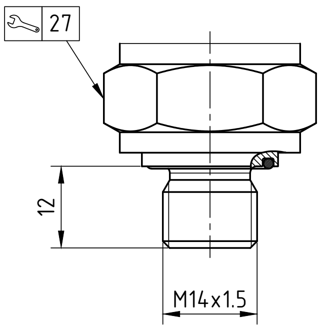
M20x1.5 EN 837; code 5
G1/2" EN 837; code 7



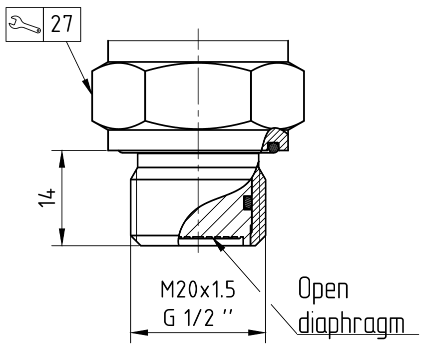
M20x1.5 DIN 3852; code 6
G1/2" DIN 3852; code 8



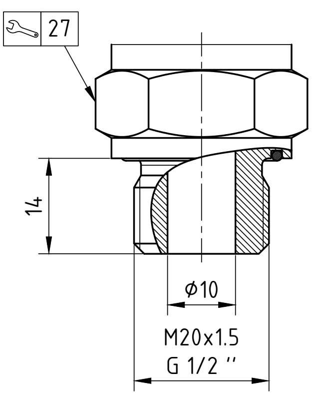
M14x1.5 DIN 3852; code 140



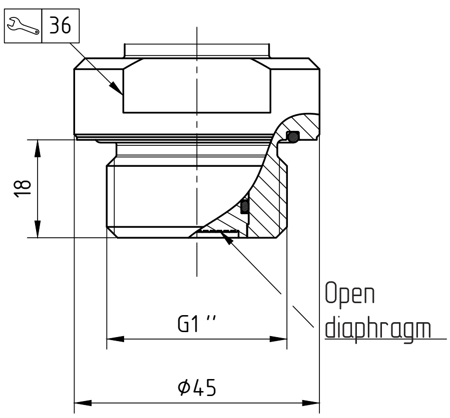
M20x1.5 DIN 3852
open diaphragm; code 205
G1/2” DIN 3852
open diaphragm; code 725



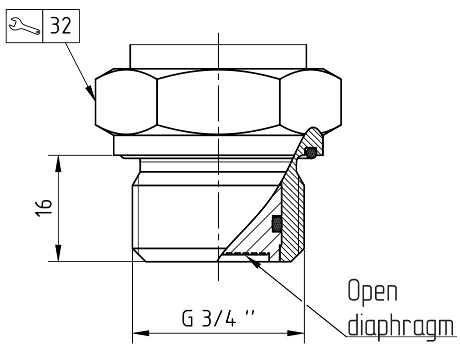
M20x1.5 DIN 3852 open diaphragm; code 206
G1/2" DIN 3852 open diaphragm; code 726



G1" DIN 3852
open diaphragm; code 705



G3/4" DIN 3852
open diaphragm; code 735



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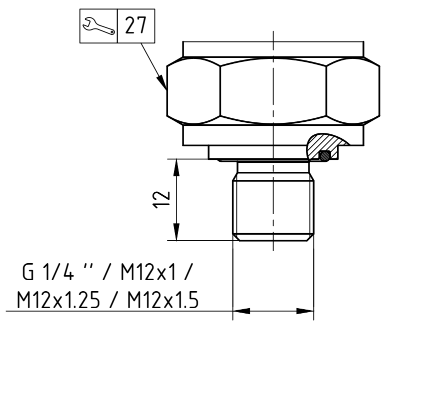
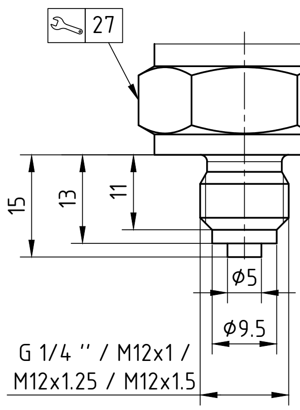
MECHANICAL CONNECTIONS FOR AMZ 5450 VERSION 0T, DIMENSIONS (mm)

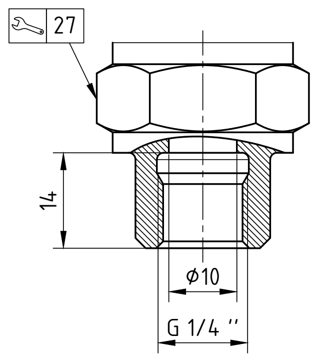
G1/4" DIN 3852; code 740
M12x1 DIN 3852; code 120
M12x1.25 DIN 3852; code 127
M12x1.5 DIN 3852; code 122



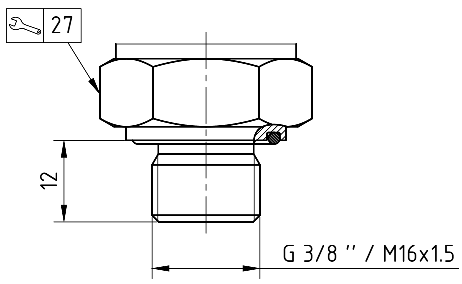
G1/4" EN 837; code 741
M12x1 EN 837; code 121
M12x1.25 EN 837; code 128
M12x1.5 EN 837; code 123



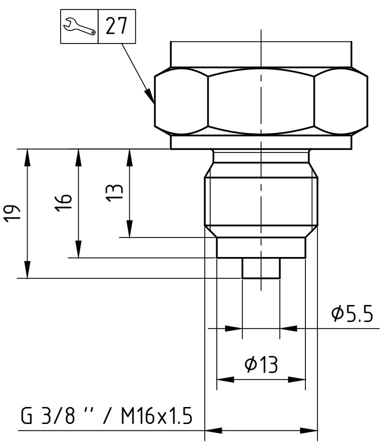
G1/4" internal thread,
10 mm through bore; code 742



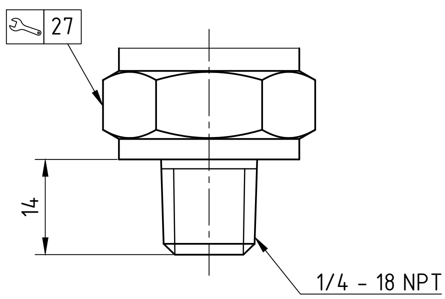
G3/8" DIN 3852; code 790
M16x1.5 DIN 3852; code 160



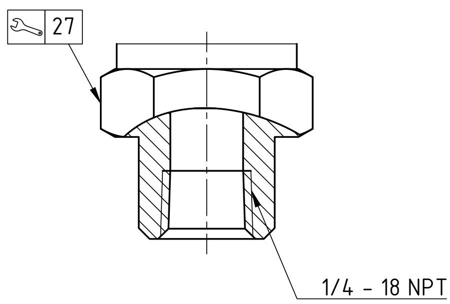
G3/8" EN 837; code 791
M16x1.5 EN 837; code 161



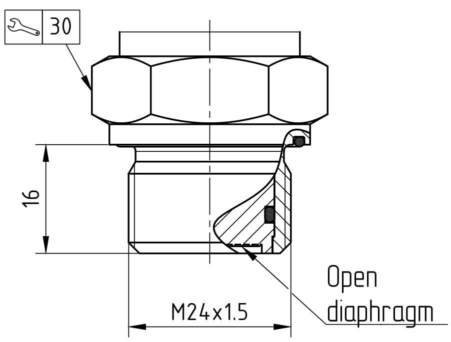
1/4" NPT; code 840



1/4" NPT internal; code 841



M24x1.5 DIN 3852 open diaphragm;
code 245



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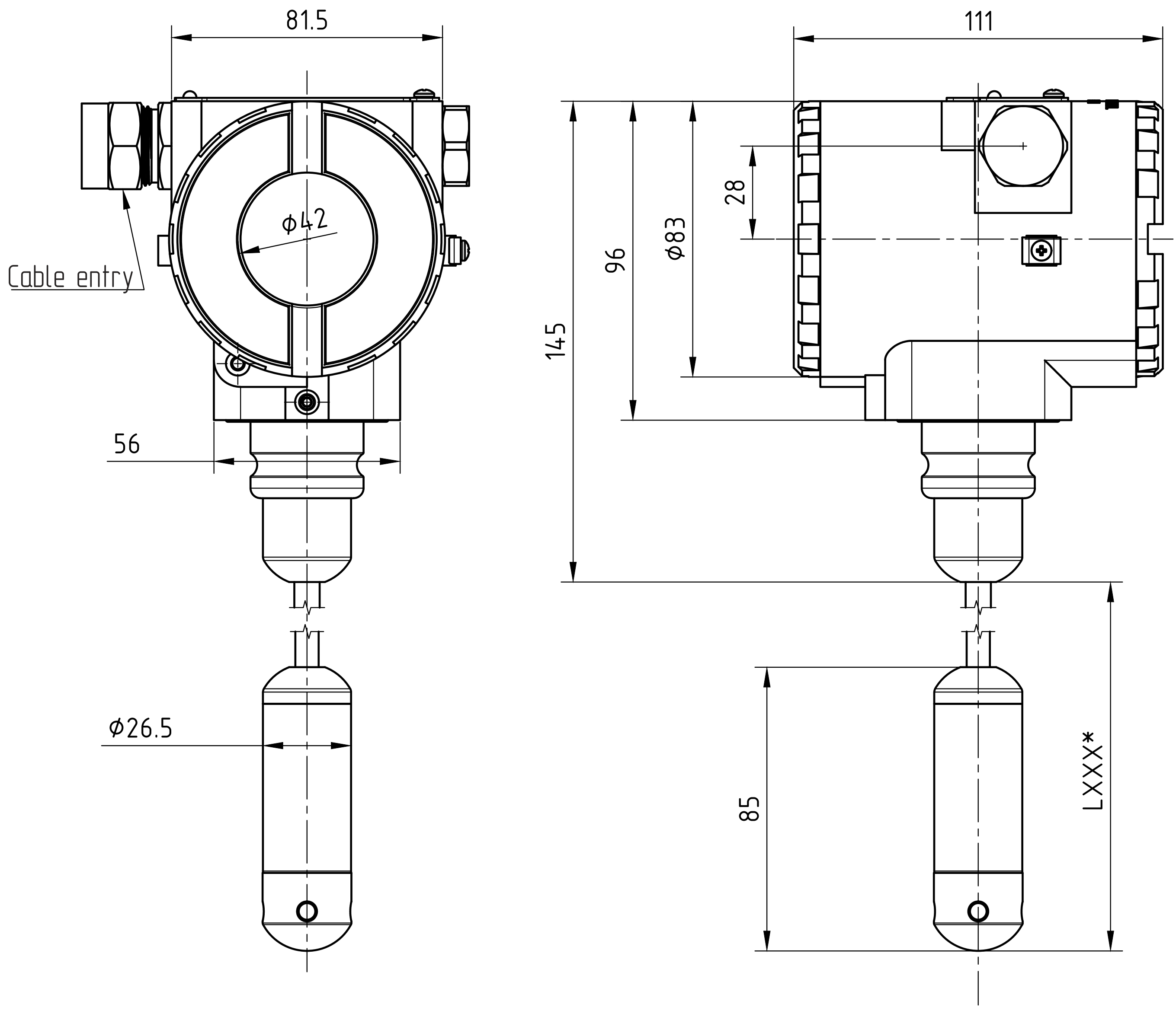
DIMENSIONS (mm) ALUMINUM ALLOY HOUSING

AMZ 5450 0T

Version with hydrostatic submersible pressure sensor

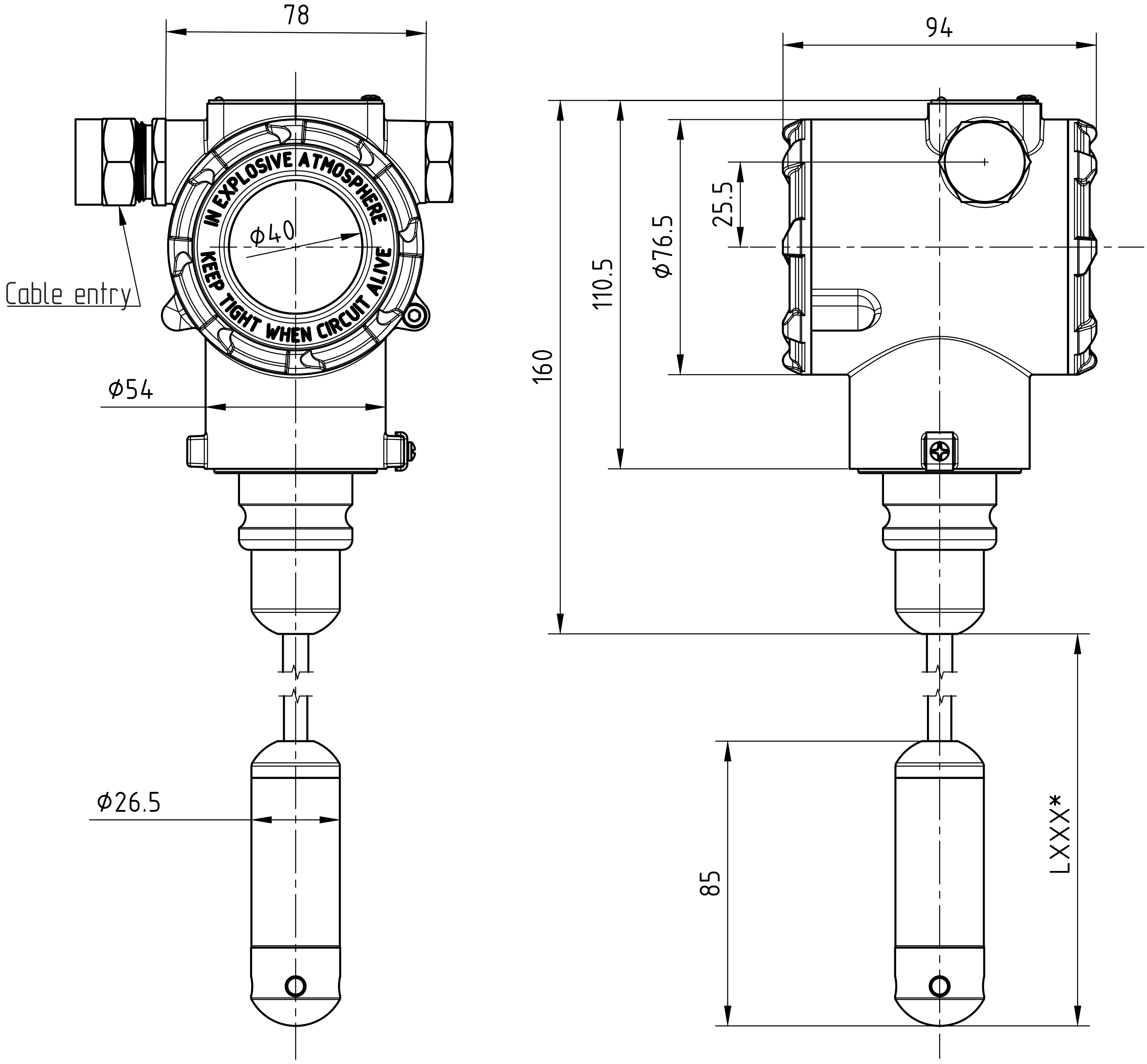


* - The length and material of the cable jacket are agreed upon when ordering and are included in its code.

DIMENSIONS (mm) STAINLESS STEEL HOUSING

AMZ 5450 0T-SS

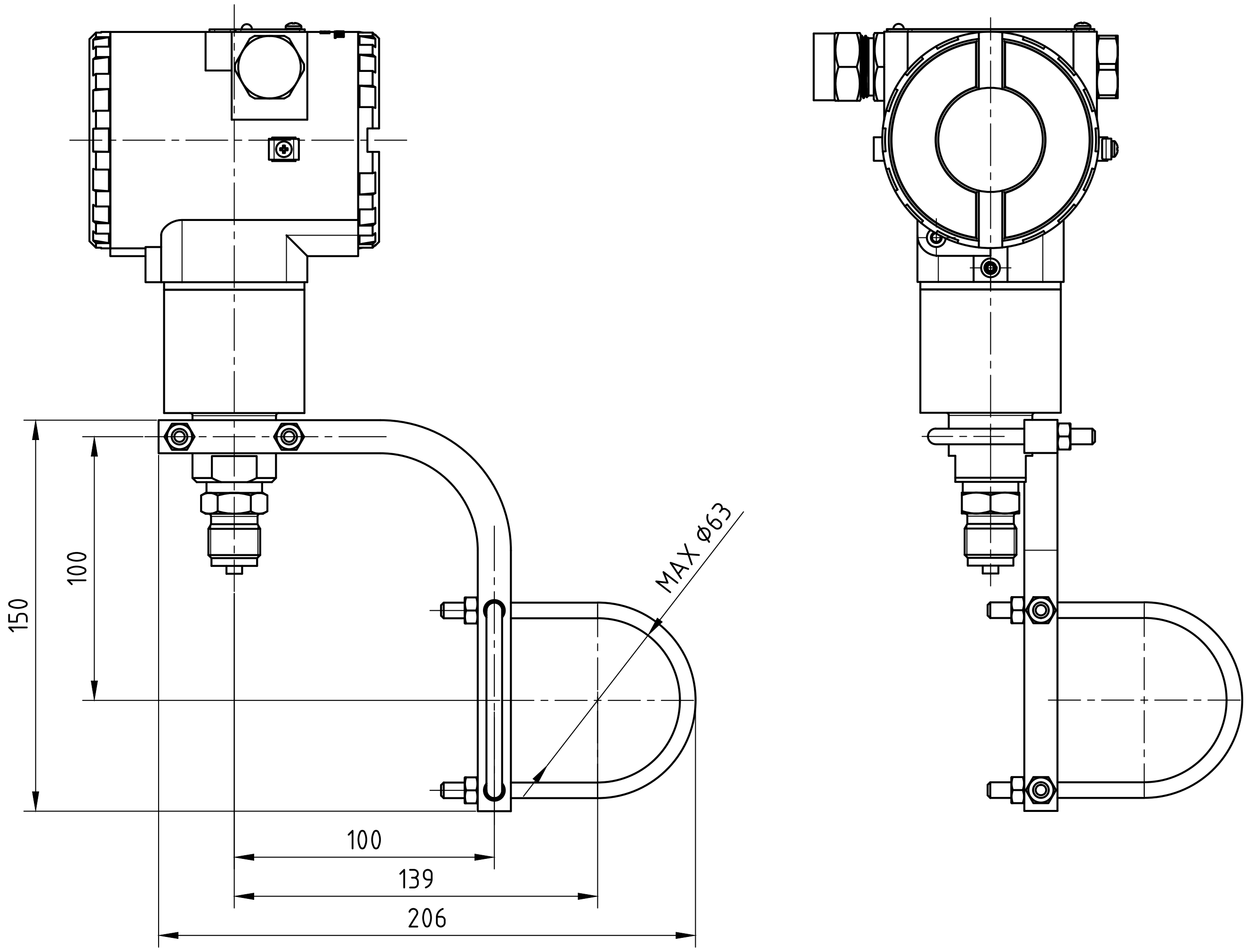
Version with hydrostatic submersible pressure sensor



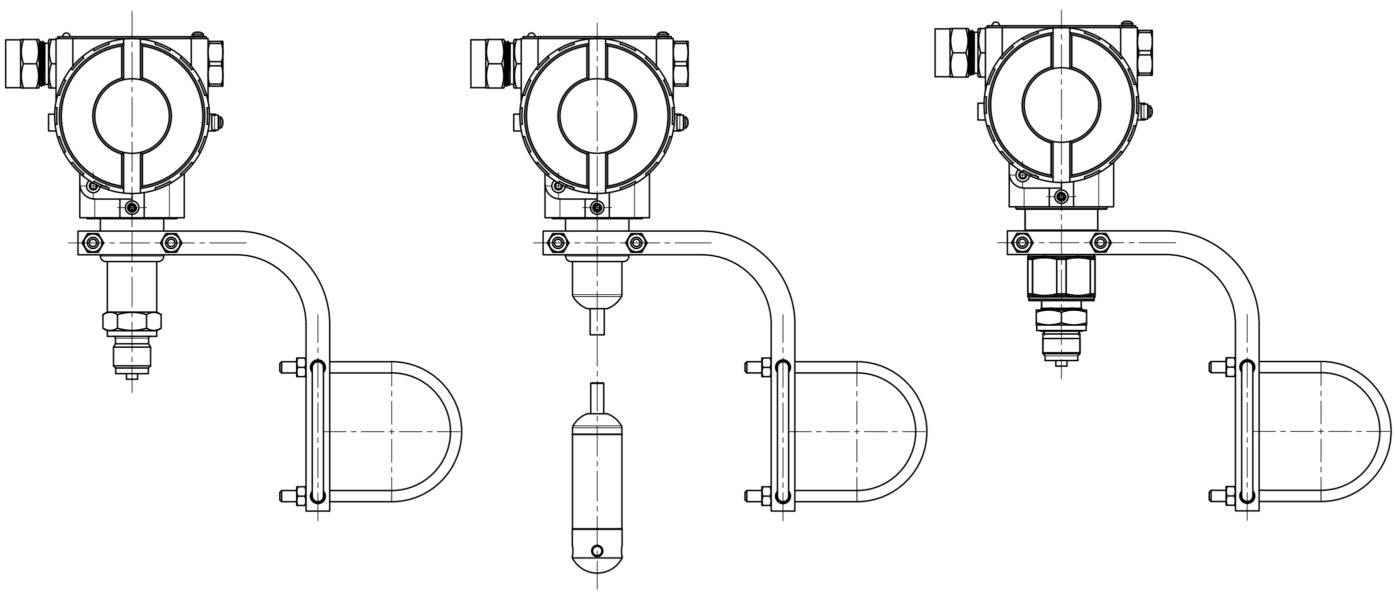
* - The length and material of the cable jacket are agreed upon when ordering and are included in its code.

DIMENSIONS (mm) ALUMINUM ALLOY HOUSING

Pipe mounting of AMZ 5450 on bracket, code 1

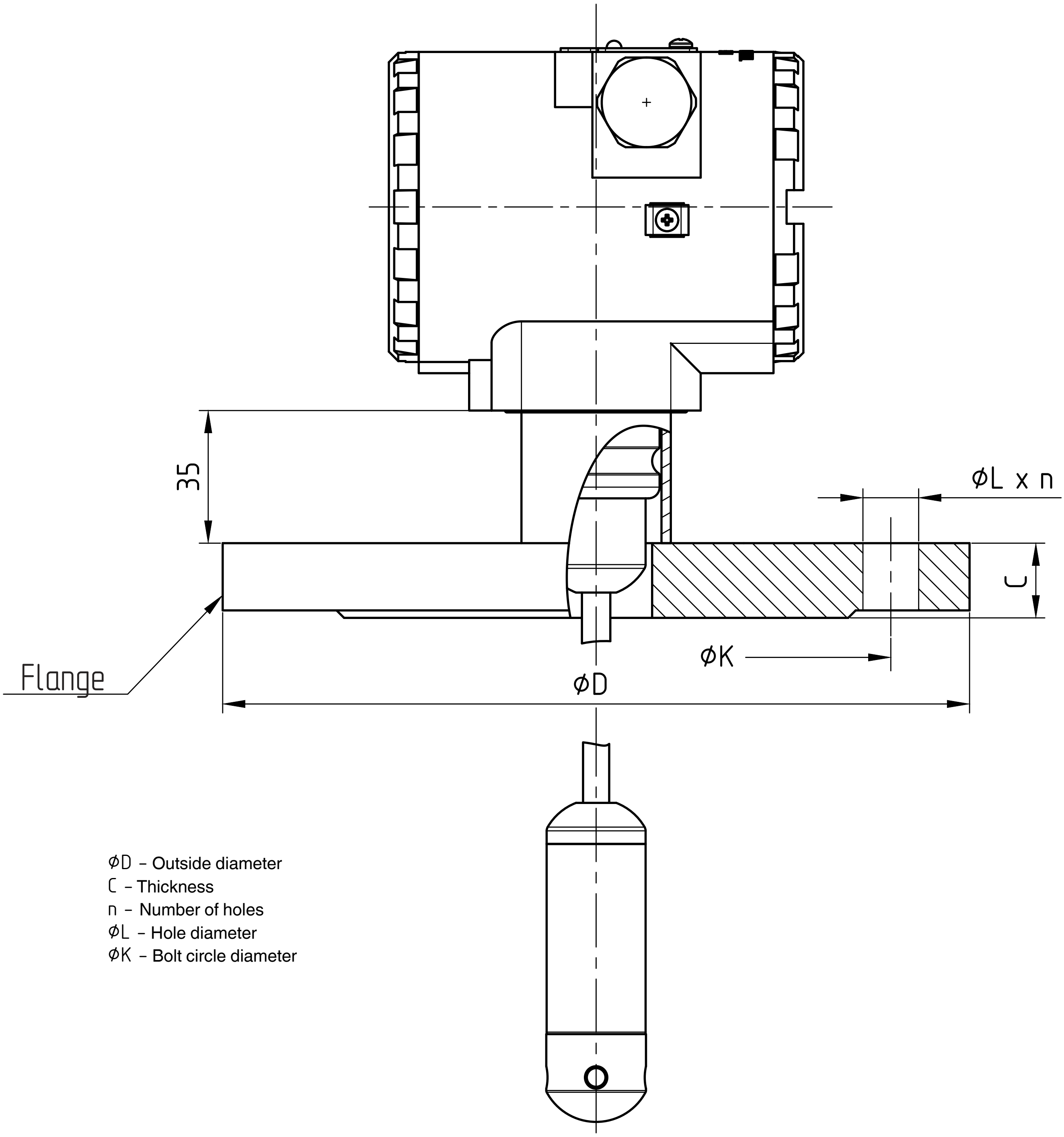


AMZ 5450 pipe mounting with bracket – various versions



DIMENSIONS (mm) ALUMINUM ALLOY HOUSING

Flange-mounted hydrostatic submersible pressure sensor version



ØD – Outside diameter
C – Thickness
n – Number of holes
ØL – Hole diameter
ØK – Bolt circle diameter

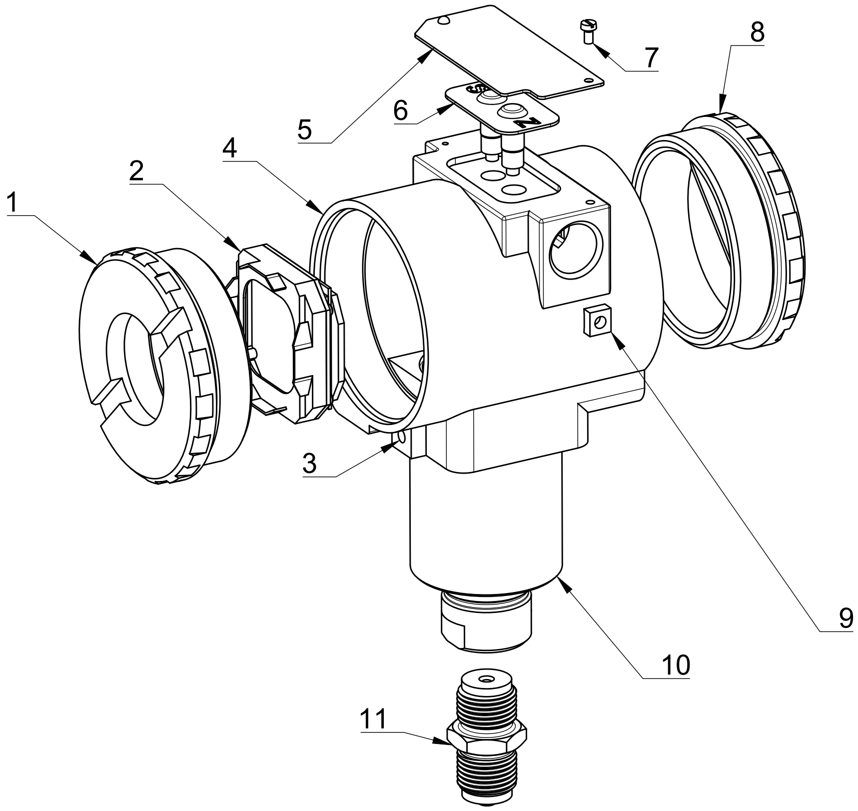
DN	PN	ØD	ØK	ØL	n	C	Bolts	Code X
DN 50	PN 10-40	165	125	18	4	18	4xM16	Code 5
DN 80	PN 10-16	200	160	18	8	20	8xM16	Code 6
DN 100	PN 10-16	220	180	18	8	20	8xM16	Code 4

Other mounting flange sizes are available upon request.

Cable jacket material and length are agreed upon order and included in the product code.

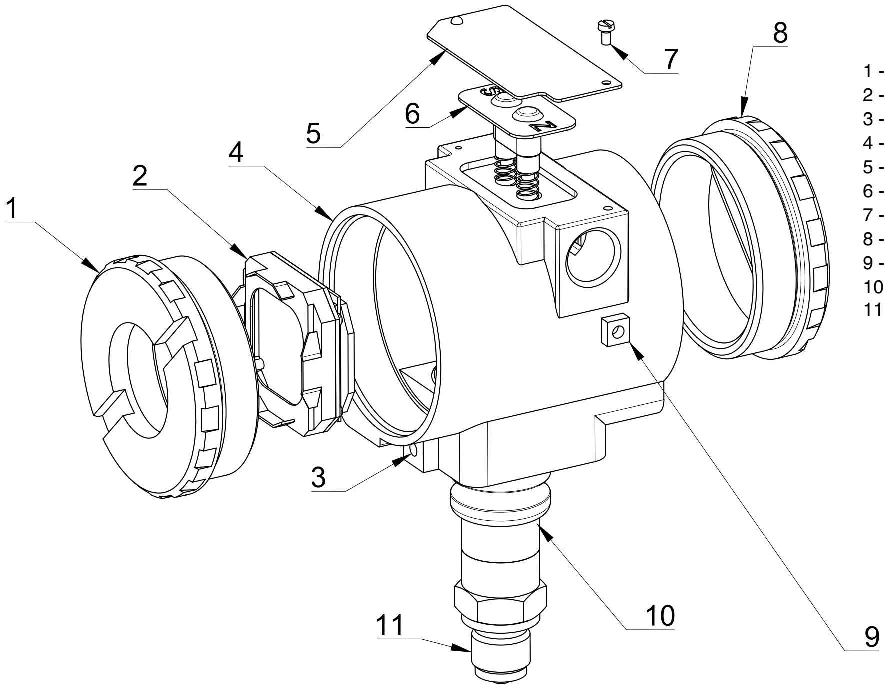
COMPONENTS

AMZ 5450 Version 00



1 - Protective display cover
2 - Display (turnable in 90°)
3 - Locking screw (unscrew to turn the housing)
4 - Housing
5 - Local adjustment button cover
6 - Setting buttons
7 - Cover screw
8 - Terminal board cover
9 - Threaded hole for housing ground screw
10 - Sensor housing
11 - Adapter

AMZ 5450 Version 0T



1 - Protective display cover
2 - Display (turnable in 90°)
3 - Locking screw (unscrew to turn the housing)
4 - Housing
5 - Local adjustment button cover
6 - Setting buttons
7 - Cover screw
8 - Terminal board cover
9 - Threaded hole for housing ground screw
10 - Sensor housing
11 - Threaded process connection

PIEZUS

ORDERING CODE

AMZ 5450		-X	-XXXX	-XX	-X	-X	-X	-X	-XXXX	-X	-XXXXX	-X	-X	-XX
PRESSURE TYPE														
Gauge pressure		G												
Absolute pressure		A												
UPPER RANGE LIMIT (URL)														
Gauge pressure	Absolute pressure													
1.5 kPa	–		1500											
7.5 kPa	–		7500											
37 kPa	37 kPa		3701											
187 kPa	187 kPa		1872											
690 kPa	690 kPa		6902											
2 MPa	2 MPa		2003											
7 MPa	7 MPa		7003											
20 MPa	20 MPa		2004											
40 MPa	40 MPa		4004											
60 MPa	60 MPa		6004											
Others	Others		XXXX											
DIAPHRAGM MATERIAL / FILL FLUID														
Stainless steel 316L / Silicone oil				11										
PROCESS CONNECTION MATERIAL														
Stainless steel 316L					S									
SEALS														
No seal (standard)						W								
FKM (fluororubber), DIN 3852						F								
NBR (nitrile butadine rubber), DIN 3852						N								
PTFE (polytetrafluoroethylene), DIN 3852						P								
EPDM (ethylene propylene rubber), DIN 3852						E								
ACCURACY														
0.075% (7.5 kPa ≤ P ≤ 7 MPa, Version 00)							Z							
0.1% (Version 0T, Version 00 with URL = 1.5 kPa)							A							
0.15%							G							
DISPLAY														
None								0						
Yes								1						
Yes / With buttons								2						
ELECTRICAL CONNECTION														
Appendix A - Cable connections*														
OUTPUT SIGNAL														
4…20 mA / HART®										H				
4…20 mA / HART® / 0Ex ia IIC T6...T4 Ga X										I				
4…20 mA / HART® / 1Ex d ia IIC T6...T4 Gb X										W				
4…20 mA / HART® / 1Ex d IIC T6...T4 Gb X										P				
MECHANICAL CONNECTION														
1/2" NPT, external											1			
1/2" – 14 NPT, internal (standard)											2			
M20×1.5 EN 837, external											5			
M20×1.5 DIN 3852, external											6			
G1/2" EN 837, external											7			
G1/2" DIN 3852, external											8			
Threaded with welding nipple: M20×1.5 EN 837, nipple + union nut included											51			
Threaded with welding nipple: M20×1.5 EN 837, 14 mm ext. nipple + union nut											52			
Threaded with welding nipple: M20×1.5 EN 837, 12 mm ext. nipple + union nut											53			

*** – The number of possible electrical connection configurations is not limited to the list presented in Appendix A; the required configuration can be agreed upon at the time of ordering.**

Continued on next page

PIEZUS

ORDERING CODE (CONTINUED)

AMZ 5450 -X -XXXX -XX -X -X -X -X -XXXX -X	-XXXXX	-X	-X	-XX
MECHANICAL CONNECTION (CONTINUED)				
M12×1 DIN 3852, external	120			
M12×1 EN 837, external	121			
M12×1.5 DIN 3852, external	122			
M12×1.5 EN 837, external	123			
M12×1.25 DIN 3852, external	127			
M12×1.25 EN 837, external	128			
M14×1.5 DIN 3852, external	140			
M16×1.5 DIN 3852, external	160			
M16×1.5 EN 837, external	161			
M20×1.5 EN 837, with built-in pressure surge damper	202			
M20×1.5, external, open diaphragm	205			
M20×1.5, external, open port	206			
M24×1.5 DIN 3852, external, open diaphragm	245			
G1" DIN 3852, external, open diaphragm	705			
M20×1.5, external, with cooling radiator	723			
G1/2", internal	724			
G1/2" DIN 3852, external, open diaphragm	725			
G1/2" DIN 3852, external, open port	726			
G3/4" DIN, external, open diaphragm	735			
G1/4" DIN 3852, external	740			
G1/4" EN 837, external	741			
G1/4", internal, 10 mm through hole	742			
G3/8" DIN 3852, external	790			
G3/8" EN 837, external	791			
1/4" NPT, external	840			
1/4" NPT, internal	841			
Submersible sensor with PVC cable jacket	LXXXP*			
Submersible sensor with polyurethane cable jacket	LXXXU*			
Submersible sensor with PTFE cable jacket	LXXXF*			
VALVE MANIFOLD**				
None		0		
Included with valve manifold		1		
Supplied assembled with valve manifold***		2		
MOUNTING KITS				
Not included			0	
Bracket for 2" (50 mm) pipe or flat panel			1	
Stainless steel bracket for 2" (50 mm) pipe or flat panel			3	
Flange DN100/PN10-16, form B, ext. Ø 40 mm			4	
Flange DN50/PN10-40, form B, ext. Ø 40 mm			5	
Flange DN80/PN10-40, form B, ext. Ø 40 mm			6	
Upon request			X	
VERSION				
0...1,5 kPa - 0...7 MPa				00
0...37 kPa - 0...60 MPa				0T
Oxygen Version (only FKM sealing and URL ≤ 170 bar)				DG
Stainless steel housing 316L				SS

Example: AMZ 5450 G-7003-11-S-W-A-1-M01-I-2-0-2-SS

* – Example: L010P - L010P – hydrostatic version: submersible sensor, depth 10 meters, PVC cable jacket.
** – Specify as a separate item when ordering.
*** – The transmitter is supplied complete with a valve assembly. A leak test is performed after installation.

APPENDIX A

Order Code	Thread of Electrical Connection	Material	Thread of Cable Entry	For Cable, mm	Ø Outer Diameter of Armor, mm	Nominal Metal Hose Diameter, mm	Protection Class (GOST 14254)	Explosion Protection	Note
Without cable entry									
M00	Internal M20x1.5	-	-	-	-	-	-	-	With plastic plugs, without cable entry
M02S	Internal M20x1.5	-	-	-	-	-	-	-	With explosion-proof plugs made of stainless steel, IP66-68, without cable entry
N00	Internal 1/2" NPT	-	-	-	-	-	-	-	With plastic plugs, without cable entry
N02S	Internal 1/2" NPT	-	-	-	-	-	-	-	With explosion-proof plugs, made of stainless steel, IP66-68, without cable entry
Cable entries with M20x1.5 threads									
M01	Internal M20x1.5	Nickel-plated brass	External M20x1.5	6-12	-	-	IP66-68	General purpose Exi, Exd	-
M03	Internal M20x1.5	Nickel-plated brass	External M20x1.5	6-12	9-17	-	IP66-68	General purpose Exi, Exd	For armored cable
M04	Internal M20x1.5	Nickel-plated brass	External M20x1.5	6-12	-	15	IP66-68	General purpose Exi, Exd	For unarmored cable with the option of metal hose connection
M05	Internal M20x1.5	Nickel-plated brass	External M20x1.5	6-14	-	20	IP66-68	General purpose Exi, Exd	For unarmored cable with the option of metal hose connection
M06	Internal M20x1.5	Nickel-plated brass	External M20x1.5	6-12	-	-	IP66-68	General purpose Exi, Exd	For unarmored cable with an adapter for metal hose with internal thread G 1/2"
M08	Internal M20x1.5	Nickel-plated brass	External M20x1.5	6-14	-	18	IP66-68	General purpose Exi, Exd	With an adapter for metal hose (RZ-TsKh-18/MRPI-18)
M10	Internal M20x1.5	Nickel-plated brass	External M20x1.5	6-14	-	15	IP66-68	General purpose Exi, Exd	-
M14	Internal M20x1.5	Nickel-plated brass	External M20x1.5	6-14	-	20	IP66-68	General purpose Exi, Exd	-
M01S	Internal M20x1.5	Stainless steel	External M20x1.5	6-12	-	-	IP66-68	General purpose Exi, Exd	-
M03S	Internal M20x1.5	Stainless steel	External M20x1.5	6-12	9-17	-	IP66-68	General purpose Exi, Exd	For armored cable

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APPENDIX A (CONTINUED)

Order Code	Thread of Electrical Connection	Material	Thread of Cable Entry	For Cable, mm	Ø Outer Diameter of Armor, mm	Nominal Metal Hose Diameter, mm	Protection Class (GOST 14254)	Explosion Protection	Note
M04S	Internal M20x1.5	Stainless steel	External M20x1.5	6-12	-	15	IP66-68	General purpose Exi, Exd	For unarmored cable with the option of metal hose connection
M05S	Internal M20x1.5	Stainless steel	External M20x1.5	6-14	-	20	IP66-68	General purpose Exi, Exd	For unarmored cable with the option of metal hose connection
M06S	Internal M20x1.5	Stainless steel	External M20x1.5	6-12	-	-	IP66-68	General purpose Exi, Exd	For unarmored cable with an adapter for metal hose with internal thread G 1/2"
M08S	Internal M20x1.5	Stainless steel	External M20x1.5	6-14	-	18	IP66-68	General purpose Exi, Exd	With an adapter for metal hose
M10S	Internal M20x1.5	Stainless steel	external M20x1.5	6-14	-	15	IP66-68	General purpose Exi, Exd	-
M14S	Internal M20x1.5	Stainless steel	External M20x1.5	6-14	-	20	IP66-68	General purpose Exi, Exd	-
M01P	Internal M20x1.5	Polyamide	External M20x1.5	6-12	-	-	-	General purpose	-
Cable entries with 1/2" NPT threads									
N01	Internal 1/2" NPT	Nickel-plated brass	external 1/2" NPT	6-12	-	-	IP66-68	General purpose Exi, Exd	-
N01P	Internal 1/2" NPT	Polyamide	External 1/2" NPT	6-12	-	-	IP66-68	General purpose	-
N03	Internal 1/2" NPT	Nickel-plated brass	External 1/2" NPT	6-12	9-17	-	IP66-68	General purpose Exi, Exd	For armored cable for unarmored cable with the option of metal hose connection
N04	Internal 1/2" NPT	Nickel-plated brass	External 1/2" NPT	6-12	-	15	IP66-68	General purpose Exi, Exd	For unarmored cable with the option of metal hose connection
N05	Internal 1/2" NPT	Nickel-plated brass	External 1/2" NPT	6-14	-	20	IP66-68	General purpose Exi, Exd	For unarmored cable with an adapter for metal hose with internal thread G 1/2"
N06	Internal 1/2" NPT	Nickel-plated brass	External 1/2" NPT	6-12	-	-	IP66-68	General purpose Exi, Exd	-
N01S	Internal 1/2" NPT	Stainless steel	External 1/2" NPT	6-12	-	-	IP66-68	General purpose Exi, Exd	For armored cable
N03S	Internal 1/2" NPT	Stainless steel	External 1/2" NPT	6-12	9-17	-	IP66-68	General purpose Exi, Exd	For unarmored cable with the option of metal hose connection
N04S	Internal 1/2" NPT	Stainless steel	External 1/2" NPT	6-12	-	15	IP66-68	General purpose Exi, Exd	For unarmored cable with the option of metal hose connection

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PIEZUS

APPENDIX A (CONTINUED)

Order Code	Thread of Electrical Connection	Material	Thread of Cable Entry	For Cable, mm	Ø Outer Diameter of Armor, mm	Nominal Metal Hose Diameter, mm	Protection Class (GOST 14254)	Explosion Protection	Note
N05S	Internal 1/2" NPT	Stainless steel	External 1/2" NPT	6-14	-	20	IP66-68	General purpose Exi, Exd	for unarmored cable with the option of metal hose connection
N06S	Internal 1/2" NPT	Stainless steel	External 1/2" NPT	6-12	-	-	IP66-68	General purpose Exi, Exd	for unarmored cable with an adapter for metal hose with internal thread G 1/2"
Other electrical connections									
R	Electrical connector 2PMG14B4SH1E2B (socket 2PM14KPN4G181)								
D	DIN 43650A, includes plug and socket								

Note: Possible electrical connection configurations are not limited to the list provided in Appendix A; the required configuration can be agreed upon when placing the order.